

HW 5

1a. $\sum_{i=1}^n x_i = 7$, purchase

$$n=15 \quad n_1 = 7 \quad n_0 = 8$$

$$D = \{1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 0, 1, 0\}$$

$$p \sim \text{Beta}(\alpha_1, \alpha_0) \quad \alpha_1 = 3, \alpha_0 = 4$$

$$L(p) = P(D|p) = p^7(1-p)^8$$

$$b. \ell(p) = 7 \ln p + 8 \ln(1-p)$$

$$\frac{d\ell(p)}{dp} = \frac{7}{p} - \frac{8}{1-p} = 0 \quad \frac{7}{p} = \frac{8}{1-p}$$

$$7(1-p) = 8p \quad 7 - 7p = 8p \quad 7 = 15p$$

$$\hat{p}_{MLE} = \frac{7}{15}$$

$$c. \hat{p}(p) = \text{Beta}(p|\alpha_1, \alpha_0) = c p^{\alpha_1-1} (1-p)^{\alpha_0-1}$$

$$= \text{Beta}(p|\alpha_1, \alpha_0) = c p^2 (1-p)^3$$

$$P(p|D) = \text{Beta}(p|N_1 + \alpha_1, N_0 + \alpha_0) = \text{Beta}(10, 12)$$

$$d. \hat{p}_{BE} = \frac{N_1 + \alpha_1}{N_1 + N_0 + \alpha_1 + \alpha_0} = \frac{10}{22} = \frac{5}{11}$$

$$e. \hat{p}_{MLE} = \frac{7}{15} = 0.47 \quad \hat{p}_{BE} = \frac{5}{11} = 0.45$$

$$\text{prior} \rightarrow \text{Beta}(\alpha_1, \alpha_0) \quad \frac{\alpha_1}{\alpha_1 + \alpha_0} = \frac{3}{7} = 0.43$$

The two estimations are very similar with $\hat{p}_{BE}$ being slightly lower. The prior affects the estimate by making the estimates closer to it, which is why the Bayesian is closer to it. With a lower sample size the Bayesian estimator is less extreme because it is only based off the sample size and is effected by prior information.

$$|V| = 11 \quad n_+ = 12 \quad n_- = 12$$

$$2a. P(+) = \frac{3}{6} = \frac{1}{2} \quad P(-) = \frac{3}{6} = \frac{1}{2}$$

$$n_k=3 \quad b. P(\text{great}|+) = \frac{3+1}{12+11} = \frac{4}{23}$$

$$n_k=3 \quad P(\text{fun}|+) = \frac{3+1}{12+11} = \frac{4}{23}$$

$$n_k=1 \quad P(\text{moe}|+) = \frac{1+1}{12+11} = \frac{2}{23}$$

$$n_k=0 \quad P(\text{great}|-) = \frac{0+1}{12+11} = \frac{1}{23}$$

$$n_k=0 \quad P(\text{fun}|-) = \frac{0+1}{12+11} = \frac{1}{23}$$

$$n_k=1 \quad P(\text{moe}|-) = \frac{1+1}{12+11} = \frac{2}{23}$$

$$c. P(+) P(d|+) = P(+) \cdot P(\text{great}|+) \cdot P(\text{fun}|+) \cdot$$

$$P(\text{moe}|+) = \frac{1}{2} \cdot \frac{4}{23} \cdot \frac{4}{23} \cdot \frac{2}{23} = 0.0013$$

$$P(-) P(d|-) = P(-) \cdot P(\text{great}|-) \cdot P(\text{fun}|-) \cdot$$

$$P(\text{moe}|-) = \frac{1}{2} \cdot \frac{1}{23} \cdot \frac{1}{23} \cdot \frac{2}{23} = 8.219 \cdot 10^{-5}$$

$$d. \text{Positive b/c } P(+) P(d|+) > P(-) P(d|-)$$

e. Bag-of-words means each doc is treated as a combination where order does not matter; thus the words are like independent factors that determine the class. This method works in practice because words are correlated to + and - connotations, which makes this method easy in practice.

HW 5

3a. In Python

b. In Python

c. In Python

d. In Python

e. One pattern is that for every feature but sepal width the 'size' from smallest to largest is setosa, versicolor, virginica.